# Even Wachs permutation polytopes are products

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**Abstract**

The convex hull of the Wachs permutations in $S_{2m}$ is affinely isomorphic to the product of the ordinary $(m-1)$-dimensional permutahedron and an $m$-cube. In particular it is simple for every $m \geq 1$, proving Conjecture 6.3 of Brenti and Sentinelli.

## Task 1: the source conjecture

For $i \in [2m]$, put $i^* = i+1$ if $i$ is odd and $i^* = i-1$ if $i$ is even. The source definition is

$$\mathcal{W}(S_{2m}) = \{w \in S_{2m} : |w^{-1}(i) - w^{-1}(i^*)| \leq 1 \text{ for every } i \in [2m-1]\}.$$

We identify $w$ with $(w(1), \ldots, w(2m)) \in \mathbb{R}^{2m}$ and set $P_m = \operatorname{conv} \mathcal{W}(S_{2m})$. Conjecture 6.3 of [1] asks whether $P_m$ is simple for every positive integer $m$. Simplicity is measured in the affine hull: a $d$-dimensional polytope is simple if each vertex is incident with exactly $d$ edges.

**Theorem 1.** *Let*

$$\Pi_m = \operatorname{conv}\{(\sigma(1), \ldots, \sigma(m)) : \sigma \in S_m\}, \qquad C_m = [-1,1]^m.$$

*Then $P_m$ is affinely isomorphic to $\Pi_m \times C_m$. It has dimension $2m-1$, has $2^m m!$ vertices, and is simple. Consequently its number of edges is $(2m-1)2^{m-1}m!$.*

**Lemma 1.** *Every $w \in \mathcal{W}(S_{2m})$ is uniquely obtained by arranging the $m$ pairs $\{2j-1, 2j\}$ in an arbitrary order and choosing the order inside each pair independently.*

*Proof.* The positions occupied by $2j-1$ and $2j$ are distinct and have distance at most one, so they are consecutive. These disjoint position pairs cover $[2m]$. Position 1 must be paired with position 2; removing them and repeating shows that the position pairs are precisely $\{1,2\}, \{3,4\}, \ldots, \{2m-1, 2m\}$. The value pair placed in each position pair is arbitrary, as is its orientation. Conversely, every such arrangement satisfies every condition in the definition. □

## The affine product

For $x \in \mathbb{R}^{2m}$ define $T(x) = (y, z) \in \mathbb{R}^m \times \mathbb{R}^m$ by

$$y_i = \frac{x_{2i-1} + x_{2i} + 1}{4}, \qquad z_i = x_{2i} - x_{2i-1} \quad (1 \leq i \leq m). \tag{1}$$

This affine map is invertible, with

$$x_{2i-1} = 2y_i - \frac{1}{2} - \frac{z_i}{2}, \qquad x_{2i} = 2y_i - \frac{1}{2} + \frac{z_i}{2}. \tag{2}$$

By Lemma 1, on the Wachs set the vector $y$ runs over all permutations of $(1,\dots,m)$ and $z$ runs independently over $\{-1,1\}^m$. Hence

$$T(\mathcal{W}(S_{2m})) = S_m \times \{-1,1\}^m,$$

where $S_m$ on the right denotes its one-line coordinate vectors. For finite sets $A, B$ one has $\mathrm{conv}(A \times B) = \mathrm{conv}(A) \times \mathrm{conv}(B)$: the nontrivial inclusion follows by multiplying convex-combination weights, $(\sum_a \lambda_a a, \sum_b \mu_b b) = \sum_{a,b} \lambda_a \mu_b (a,b)$. Affine maps preserve convex combinations, proving

$$T(P_m) = \Pi_m \times C_m. \tag{3}$$

## Dimension and simplicity, with an explicit cone proof

Write $v = (1,\dots,m)$ and $g_k = e_k - e_{k+1}$ for $1 \le k < m$. For a permutation vector $u$, define

$$a_k = \sum_{i=1}^{k} (u_i - i) \ge 0.$$

The inequality holds because any $k$ distinct entries of $[m]$ have sum at least $1 + \cdots + k$. Since $\sum_i (u_i - i) = 0$, telescoping gives

$$u - v = \sum_{k=1}^{m-1} a_k g_k. \tag{4}$$

Each $g_k$ itself is a vertex difference, obtained by interchanging the entries $k, k+1$ in $v$. These $m-1$ vectors are linearly independent. Thus the cone of feasible directions of $\Pi_m$ at $v$ is exactly their nonnegative span and has dimension $m-1$. Coordinate permutations act transitively on the permutation vectors. Each vector is a vertex: the functional $u \mapsto v \cdot u$ is uniquely maximized at $v$, since

$$v \cdot u = \|v\|^2 - \tfrac{1}{2}\|u - v\|^2$$

for permutation vectors $u$, and the same argument applies after permuting coordinates.

At the vertex $(v, \mathbf{1})$ of the product in (3), the cone of feasible directions is generated by

$$(g_k, 0)\ (1 \le k < m), \qquad (0, -2e_i)\ (1 \le i \le m).$$

Indeed, (4) supplies the first components, while each cube vertex $z$ satisfies $z - \mathbf{1} = \sum_i (1 - z_i)(-2e_i)/2$. Every displayed generator is itself a vertex difference. The $2m-1$ generators are linearly independent, so this cone has exactly $2m-1$ extreme rays. Extreme rays of a polytope's feasible-direction cone at a vertex correspond to its incident edges: an exposing functional for such a ray cuts out the corresponding one-dimensional face, and the converse follows by exposing an edge. Consequently the product has dimension $2m-1$ and exactly $2m-1$ edges at this vertex. Coordinate permutations in the first factor and sign changes in the second act transitively on all its vertices, proving simplicity everywhere. The vertices of a product are exactly the pairs of vertices of its factors, giving $2^m m!$ of them. Counting incidences of vertices and edges proves the stated edge count. This also covers $m = 1$, when $\Pi_1$ is a point and $P_1$ is a segment, and completes the proof of Theorem 1.

## Scope and verification

The argument uses the even-index definition and proves the full positive integer range. It makes no assertion about the odd-index hulls or the Bruhat order. The independent finite verifier exhausts $S_{2m}$ for $1 \leq m \leq 4$ using the original inverse-position condition, checks the product identity and inverse map, and checks the exact cone certificates through $m = 5$. These finite checks support the universal proof above; they do not replace it. This is author-agent review, without independent peer review or a proof assistant.

## Proposed follow-up tasks

**Task 2.** Give a closed formula for every face number of $P_m$ by combining ordered set partitions for $\Pi_m$ with the faces of $C_m$.

**Task 3.** Determine the Ehrhart polynomial of $P_m$ for the lattice $\mathbb{Z}^{2m} \cap \operatorname{aff}(P_m)$, accounting for the lattice change under (1).

**Task 4.** For real numbers $a_1 < b_1 < a_2 < b_2 < \cdots < a_m < b_m$, classify when the hull of all ordered, independently oriented blocks $(a_j, b_j)$ is simple, including the case of unequal block lengths.

**Task 5.** Determine exactly which odd integers $n$ have simple $\operatorname{conv} \mathcal{W}(S_n)$; for every remaining $n$, exhibit a vertex with more incident edges than the affine dimension.

## References

[1] F. Brenti and P. Sentinelli, *Wachs permutations, Bruhat order and weak order*, European Journal of Combinatorics **119** (2024), 103804, doi:10.1016/j.ejc.2023.103804. arXiv:2212.04932v1; Conjecture 6.3, p. 36. The Polimi repository supplies a byte-identical 38-page file labeled postprint/author's accepted manuscript; its version provenance and the separately read 31-page author preprint are detailed in the accompanying literature audit.